

Task 2

$\sqrt{n}$

18

$$\sqrt{36} = 6$$

possible factors =

$$1 \times 36 = 36$$

$$2 \times 18 = 36$$

$$3 \times 12 = 36$$

$$4 \times 9 = 36$$

$$6 \times 6 = 36$$

$$\begin{array}{r} 2 \overline{) 36} \\ \underline{36} \\ 0 \end{array}$$

$$\begin{array}{r} 12 \\ 3 \overline{) 36} \\ \underline{36} \\ 0 \end{array}$$

false.

```
for (let i = 2; i <= sqrt(n); i++) {
  if (n % i === 0) {
    return false;
  }
}
```

$$n = 36$$

$$\sqrt{n} = 6$$

loop

2, 3, 4, 5, 6

Task 1

$$11 = \frac{11}{2} = 5$$

```
for (let i = 2; i <= n/2; i++) {
  if (n % i === 0) {
    return false;
  }
}
```

$$n = 11$$

$$n/2 = 5$$

loop

2, 3, 4, 5

Optimized:

1st may pahlay

if ($n \leq 1$) ga nikalay ge.
if ($n == 2$)

if ($n \% 2 == 0$) return false.
check odd.

Only
for (let $i = 3$; $i \leq \sqrt{n}$; $i += 2$)
if ($n \% i == 0$) {
return false;
}

}

$n = 37$
 $\sqrt{n} = 6$
loop
3, 5

Task 3
0.1 = 2.23
4, 6, 8... 1000
false.

outer loop Sabko list may rakh kay true
karna hai. phir saray loop
2 say chalay ge.

for (let $i = 2$; $i \leq \text{Math.sqrt}(1000)$; $i++$) {
isPrime[i] = true;

inner loop
for (let $y = i * i$; $y \leq 1000$; $y += i$) {